



Daffodil
International
University



DCCN Project Report on

Daffodil Smart City MAN (DSCM)



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Course code: **ITM 315**

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Semester: **Summer 2025**

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Submitted on 13th August 2025

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Letter of Transmittal

13th August 2025

Ms. Nishat Tasnim Shishir

Lecturer, Department of Information Technology & Management

Daffodil International University

Daffodil Smart City, Ashulia, Savar, Dhaka

Subject: Submission of Report on Daffodil Smart City MAN (DSCM).

Dear Madam,

Assalamu Alaikum. We hope you are doing well.

With due respect, we are pleased to submit our report on Daffodil Smart City MAN (DSCM) as part of our academic curriculum. We have tried our best to prepare this report using reliable information and present it in a professional way. In this report, we have analyzed real-world problems related to the topic.

We are grateful for the opportunity you have given us to work on this project. We also request you to kindly accept our report and overlook any minor mistakes that may be present due to certain limitations.

Best regards,

On behalf of Team Insight

Kazi Arafat Hossain

Declaration

We, Team Insight, Bachelor of Science in Information Technology & Management, Daffodil International University, declare that:

- 1) This project report is completely our own work, done while studying B.Sc. in ITM at Daffodil International University.
- 2) We have not copied any part from others, except where we have clearly given proper references.
- 3) This report has not been submitted before for any other degree or diploma at any university or institution.
- 4) We have mentioned and thanked all the main sources of help we received.

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Certificate of Supervisor

This is to certify that the report titled “**Daffodil Smart City MAN (DSCM)**” has been prepared by **Team Insight** under my supervision, for the academic requirement of completing the course Data Communication and Computer Network. To the best of my knowledge, this is original group work. I confirm that the project has been carried out sincerely on the stated topic under my guidance.

Nishat Tasnim Shishir

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Acknowledgement

We sincerely thank and respect everyone who helped us in preparing this report. We give special thanks to our course teacher, ***Ms. Nishat Tasnim Shishir***, for her valuable guidance and supervision. We also thank our teammates for helping to organize the sections and for giving useful suggestions to improve the report. Finally, we are grateful to all the sources from which we collected information.

Abstract

This project shows how to design and set up a three-branch network using Cisco Packet Tracer. It connects Daffodil International University (DIU), Daffodil Medical Center and Daffodil School using routers, switches, servers, and computers. Each branch has its own IP addresses, automatic IP service (DHCP), and local network. The routers are linked with serial cables so the branches can communicate and share data with each other. This design works like a real-world large network, allowing all branches to share resources and communicate quickly and easily.

Keywords: Metropolitan Area Network, MAN, Cisco Packet Tracer, Fiber Optic, Branch Connectivity.

1. Introduction

1.1 Overview of MAN

A Metropolitan Area Network (MAN) connects multiple local networks in a city or larger area. It is larger than a LAN and smaller than a WAN. MANs allow branch offices to swiftly communicate data and services across medium distances.

1.2 Problem Scenario

Daffodil Smart City (DSC) has three important branches spread across the city:

- ❖ **DIU (Main Hub):** Works as the head office and has a modern data center to store student records, financial data, medical patient files and other important information.
- ❖ **Daffodil Medical Center:** Manages official communication through email for staff, students and departments.
- ❖ **Daffodil School:** Runs the DSC internal website and student portal for announcements, academic updates and online resources.

Right now, these branches are not connected directly. Because of this:

- ❖ Staff send files by email or carry them using USB drives.
- ❖ Each branch pays for its own internet connection, which is expensive.
- ❖ Real-time communication is slow, causing delays and misunderstandings.
- ❖ Data can be duplicated or outdated because it is not shared instantly.
- ❖ This creates confusion, wasted time and extra costs.

To solve this, DSC needs a fast, secure and dependable Metropolitan Area Network (MAN) so that all branches can work together as if they were in the same building.

1.3 Problem Solution

To fix these problems, DSC will build the Daffodil Smart City Metropolitan Area Network (DSC-MAN) using high-speed fiber optic cables.

- ❖ **Main Hub (DIU):** Will host all main servers and internet connection. This includes the Database Server, which stores all important institutional data.
- ❖ **Daffodil Medical Center:** Will be connected to the MAN to access the database and also run the Email Server for official communication.
- ❖ **Daffodil School:** Will be connected to the MAN to access resources and run the Web Server for DSC's website and student portal.

How it will work:

- ❖ Each branch will connect to the DIU through secure routers and fiber links.
- ❖ All staff will be able to share files, use apps and communicate instantly.
- ❖ Only the DIU will have the main internet connection, reducing the need for separate internet packages at each branch.

Benefits:

- ❖ Saves money by removing extra internet bills.
- ❖ Increases speed and reliability of communication.
- ❖ Makes it feel like all three branches are working from one big office.
- ❖ Easy to expand in the future if DSC adds more branches.

1.4 Project Motivation

In many large cities, organizations have multiple offices in different locations. When these offices are not directly connected, they face:

- ❖ Slow file sharing.
- ❖ Poor communication.
- ❖ High internet costs.
- ❖ Data duplication and outdated information.

This is exactly what happened to DSC:

- ❖ The DIU, Daffodil Medical Center and Daffodil School could not share resources in real time.
- ❖ This caused delays in student services, staff work and overall operations.
- ❖ To solve this, DSC decided to create the DSC Metropolitan Area Network (DSC-MAN) using fiber optic technology.

With this network:

- ❖ All branches will connect to the main data center.
- ❖ They will share internet, applications and files instantly.
- ❖ Productivity will improve, costs will go down and the system can grow easily in the future.

2. Methodology

2.1 Tools used

- ❖ **Cisco Packet Tracer:** For building and testing the simulated network.
- ❖ **Router and switch CLI (IOS):** For configuration.
- ❖ **Server services in Packet Tracer:** To simulate DNS, web and email.
- ❖ **Basic documentation tools:** MS Word (report), Canva + PowerPoint (slides).

2.2 Approach

- ❖ Build topology in Packet Tracer (routers, switches, PCs, server).
- ❖ Assign IP addresses and configure routing (RIP v2).
- ❖ Enable services on server.
- ❖ Test connectivity (ping, web browser, traceroute).
- ❖ Collect screenshots and config outputs for the report.

3. Network Design

We created a Metropolitan Area Network (MAN) that connects three branches of the same organization in different parts of the city:

- ❖ DIU Main Branch (Head Office) Central hub with the main server and internet access.
- ❖ Daffodil Medical Center Branch – Focused on customer services.
- ❖ Daffodil School Branch – Handles logistics.

We used:

- ❖ 3 Routers (to connect branches over long distances)
- ❖ 3 Switches (to connect local devices inside each branch)
- ❖ 3 Servers (one for each branch)
- ❖ 9 PCs (3 in each branch)
- ❖ Fiber optic & serial connections for high-speed, secure data transfer

Topology:

- ❖ Star Topology → DIU Main Branch is in the center, directly connected to the other two branches.
- ❖ Each branch has its own local network (LAN) connected to the branch's router.
- ❖ All routers are connected using serial links so branches can share data like they're in the same building.

Why this design works well:

- ❖ Fast and reliable communication between all branches
- ❖ Centralized data access at DIU
- ❖ Cost-efficient and easy to maintain
- ❖ Secure network traffic within the organization

4. Implementation in Cisco Packet Tracer

Open Cisco Packet Tracer and save a new file: **DCCN Project DSC MAN Team Insight.pkt**. Save often. The step by step processes are described below:

1) Taking devices

- ❖ Take 3 routers (PT-router), 3 switches (2950-24) and 3 Server-PTs.
- ❖ Add 3 PCs for each branch, totaling 9 PCs.
- ❖ Switch0 will be connected to one server because it is considered the main hub (DIU).

2) Connecting switches, servers and PCs

- ❖ Connect the 3 routers to the 3 switches.
- ❖ Connect each of the 3 switches to one server.
- ❖ Connect each switch to 3 PCs using copper straight-through cables on FastEthernet ports.

3) Inter-router connections

- ❖ Connect the 3 routers to each other using serial DTE cables.
- ❖ When connecting, select Serial 2/0 for the serial interface each time.

4) Working with Router0 (DIU)

- ❖ Select Router0.
- ❖ Go to Config.
- ❖ Select FastEthernet0/0 from the left bar.
- ❖ Turn On the port status.
- ❖ Assign IP 10.168.10.100 and subnet mask 255.255.255.0.
- ❖ Close the tab.

5) Configuring Server0 (DIU)

- ❖ Select Server0.
- ❖ Go to Config.
- ❖ Set the gateway to 10.168.10.100.
- ❖ Select FastEthernet0 from the left bar.
- ❖ Assign IP 10.168.10.1 and subnet mask 255.255.255.0.
- ❖ Go to Services from the navigation bar.
- ❖ Select DHCP, turn the service On, set default gateway 10.168.10.100 and set start IP 10.168.10.2.
- ❖ Save and close the tab.

6) Configuring PCs in DIU

- ❖ Select PC0 → Desktop → IP Configuration → choose DHCP, then close the tab.
- ❖ Repeat the same for PC1 and PC2.

7) Record router IP addresses

- ❖ Note down the IP address of each router and label them under the routers in your diagram.
- ❖ IP: 10.168.10.100 for DIU, IP: 192.168.10.100 for Daffodil medical center, IP: 172.168.10.100 for Daffodil school

8) Working with Router2 (School)

- ❖ Select Router2 (School).
- ❖ Go to Config.
- ❖ Select FastEthernet0/0 from the left bar.
- ❖ Turn On the port status.
- ❖ Assign IP 172.168.10.100 and subnet mask 255.255.255.0.
- ❖ Close the tab.

9) Configuring Server2 (School)

- ❖ Select Server2.
- ❖ Go to Config.
- ❖ Set the gateway to 172.168.10.100.
- ❖ Select FastEthernet0 from the left bar.
- ❖ Assign IP 172.168.10.1 and subnet mask 255.255.255.0.
- ❖ Go to Services → DHCP, turn the service On, set default gateway 172.168.10.100, and set start IP 172.168.10.2.
- ❖ Save and close the tab.

10) Configuring PCs in School

- ❖ Select PC3 → Desktop → IP Configuration → choose DHCP, then close the tab.
- ❖ Repeat the same for PC4 and PC5.

11) Working with Router1 (Medical Center)

- ❖ Select Router1 (Medical Center).
- ❖ Go to Config.
- ❖ Select FastEthernet0/0 from the left bar.
- ❖ Turn On the port status.
- ❖ Assign IP 192.168.10.100 and subnet mask 255.255.255.0.
- ❖ Close the tab.

12) Configuring Server1 (Medical Center)

- ❖ Select Server1.
- ❖ Go to Config.
- ❖ Set the gateway to 192.168.10.100.
- ❖ Select FastEthernet0 from the left bar.
- ❖ Assign IP 192.168.10.1 and subnet mask 255.255.255.0.
- ❖ Go to Services → DHCP, turn the service On, set default gateway 192.168.10.100, and set start IP 192.168.10.2.
- ❖ Save and close the tab.

13) Configuring PCs in Medical Center

- ❖ Select PC6 → Desktop → IP Configuration → choose DHCP, then close the tab.
- ❖ Repeat the same for PC7 and PC8.

14) Local connectivity result

- ❖ Now, PCs in each branch can communicate within their own branch.
- ❖ However, they cannot yet transfer data between branches (for example, School PCs cannot send data to the Medical Center PCs).

15) Configuring Router0 serial links to other routers (DIU)

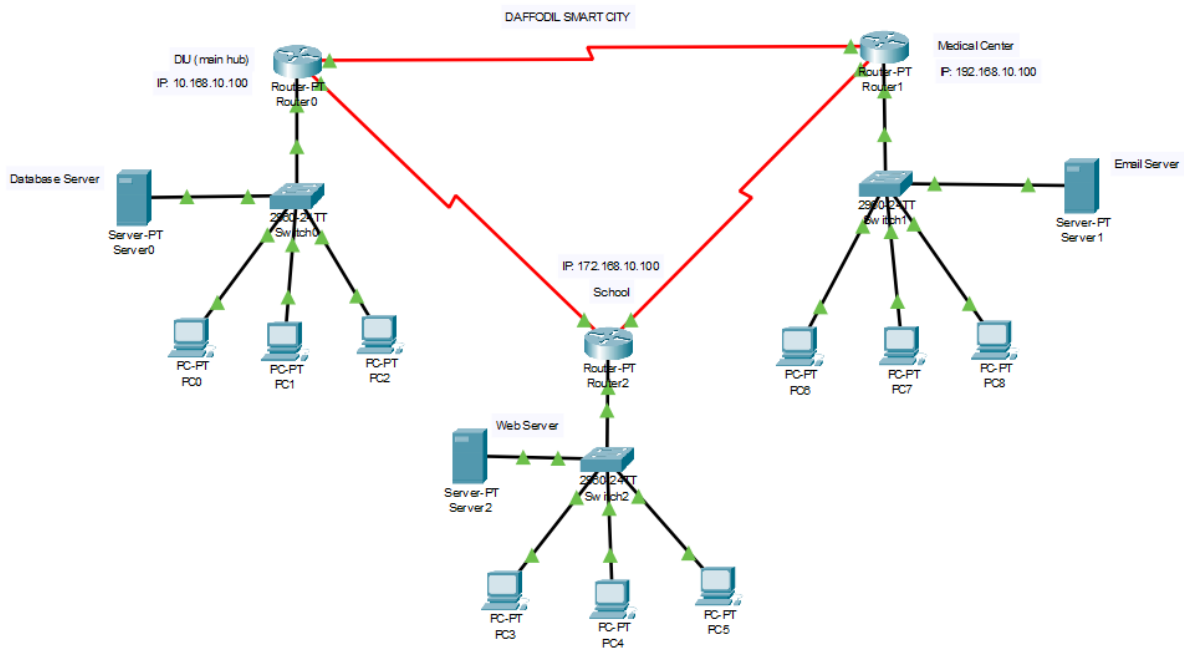
- ❖ Select Router0 (DIU main hub).
- ❖ Go to Config.
- ❖ Select Serial2/0 from the left bar, turn On the port status, set clock rate 2000000 and assign IP 172.168.10.100 with subnet mask 255.255.255.0 (this is the IP of Router2/School).
- ❖ Select Serial3/0 from the left bar, turn On the port status, and assign IP 192.168.10.100 with subnet mask 255.255.255.0 (this is the IP of Router1/Medical Center).

16) Configuring serial links on the other routers

- ❖ On Router2 (School): configure serial interfaces to include the IPs of DIU and Medical Center, and set clock rate 2000000 where needed.
- ❖ On Router1 (Medical Center): configure serial interfaces to include the IPs of DIU and School and set clock rate 2000000 where needed.

17) Inter-branch testing

- ❖ Send test data between branches, for example: PC0 → PC5, PC1 → PC2 and PC7 → PC0.
- ❖ All tests are successful, indicating that data can now be transferred between branches across the city.



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	D
	Successful	PC0	PC5	ICMP		0.000	N	0	(edit)	(d
	Successful	PC1	PC2	ICMP		0.000	N	1	(edit)	(d
	Successful	PC7	PC0	ICMP		0.000	N	2	(edit)	(d

5. Results & Testing

5.1 What to show in results section of report

- ❖ Topology screenshot (clearly labeled).
- ❖ Ping from each branch PC to server (screenshots).
- ❖ Traceroute showing traffic passes through DIU.
- ❖ show ip route proof that routers learned remote networks via RIP.
- ❖ Web browser screenshot showing server page.

5.2 Expected outcomes

- ❖ All PCs can ping each other (across branches).
- ❖ Server at DIU reachable by any branch.
- ❖ Router routing tables show networks learned by RIP.

These outcomes prove the MAN design works in simulation. Cisco Packet Tracer is used widely for such simulations.

5.3 Troubleshooting Tips

- ❖ **If ping fails:** check PC gateway and IP; check router interface IP and no shutdown.
- ❖ **If routers do not show routes:** check router rip network statements on each router.
- ❖ **If web page does not load:** make sure server has correct IP and HTTP service is ON.

Use show ip interface brief and show ip route on routers to debug.

6. Conclusion & Future Work

The Daffodil Smart City MAN (DSCM) has successfully solved the communication and data-sharing problems between our three branches in the city. By creating a star-topology Metropolitan Area Network (MAN) with the Main Branch (DIU) as the hub, all branches can now share files, internet and services quickly, safely and without delay. Our Cisco Packet Tracer simulation showed that the network works smoothly, with strong connections between all branches and central management from the main data center. This project proves that MAN technology is a smart, cost-saving and expandable solution for organizations that have multiple offices in the same city.

6.1 Limitations

- 1) **Simulation Only:** The network was built and tested in Cisco Packet Tracer, not in real life.
- 2) **No Real Internet:** The internet in the project is only simulated, not an actual working ISP connection.
- 3) **Few Devices:** We used only a small number of PCs, printers and servers for demonstration.
- 4) **Basic Security:** We did not add firewalls or intrusion detection systems in this version.

6.2 Future Work

- 1) **Real Setup:** Build the same network physically with real routers, switches and fiber optic cables.
- 2) **More Branches:** Connect more city branches without changing the main design.
- 3) **Better Security:** Add firewalls, VPNs and monitoring tools to keep the network safe from hackers.
- 4) **High Availability:** Use backup links and extra servers so the network stays online even if one part fails.
- 5) **Cloud Services:** Connect the network with cloud storage for backup, remote work and easy access from anywhere.

7. References

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